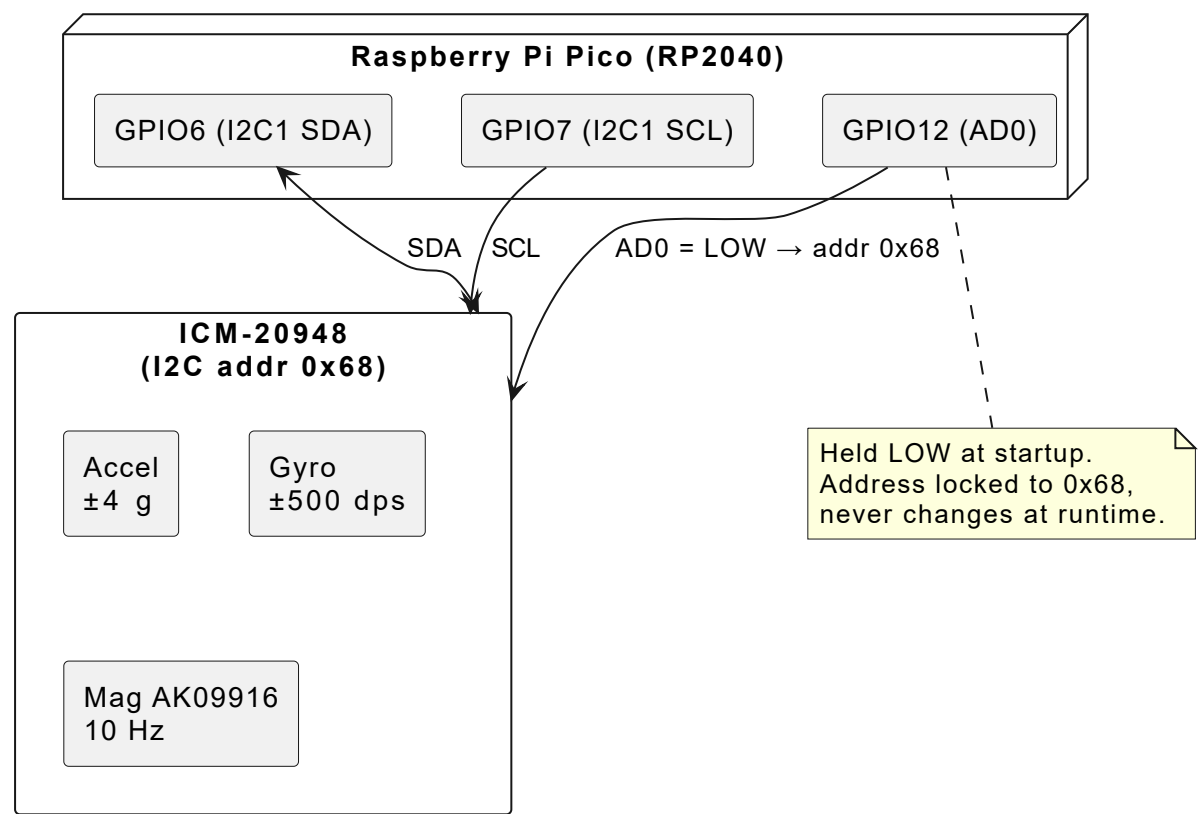


Subsystem — IMU ICM-20948

Sensor: ICM-20948 — TDK InvenSense (Adafruit #4554)
Axes: Accelerometer (3) + Gyroscope (3) + Magnetometer AK09916 (3) = 9-axis
Bus: I2C1, 400 kHz, address 0x68
Files: `include/imu.h` · `src/imu.cpp`

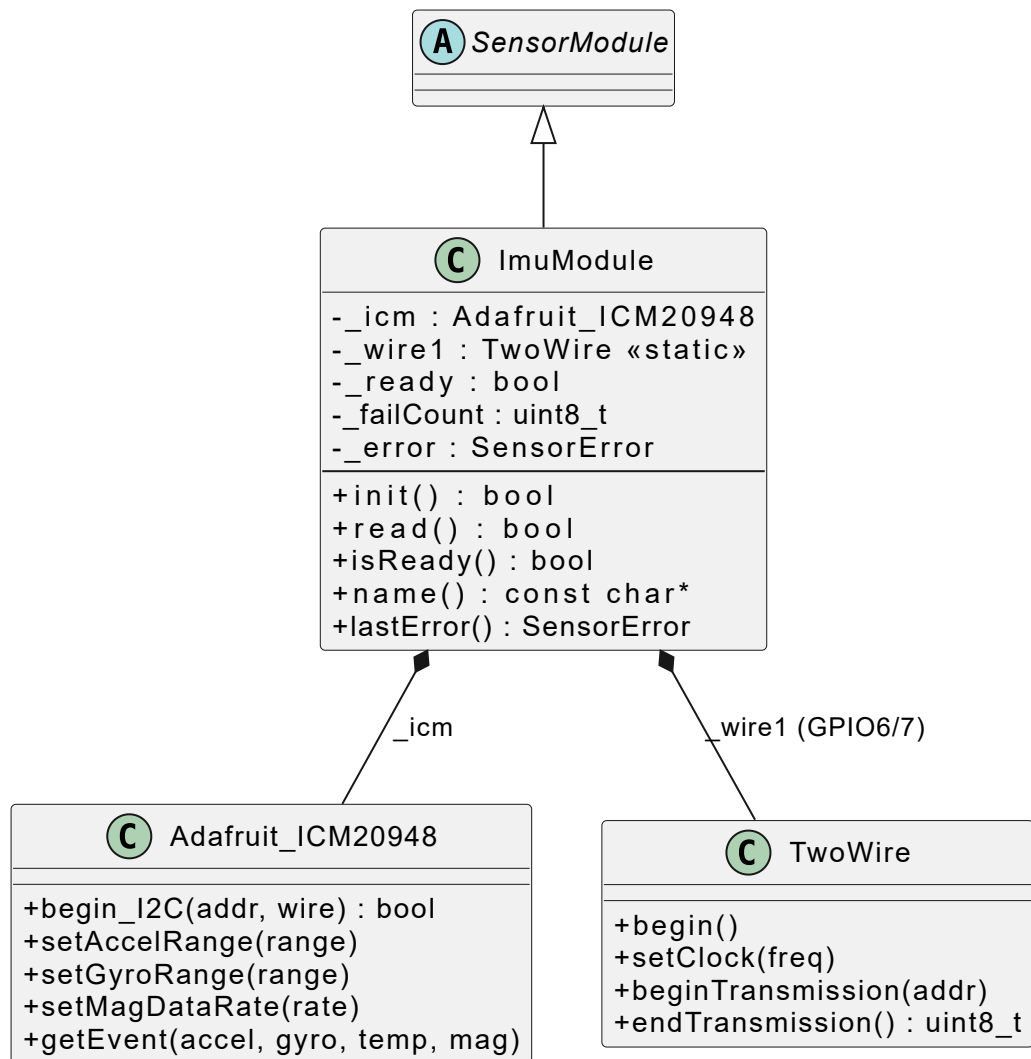
1. Hardware Connections



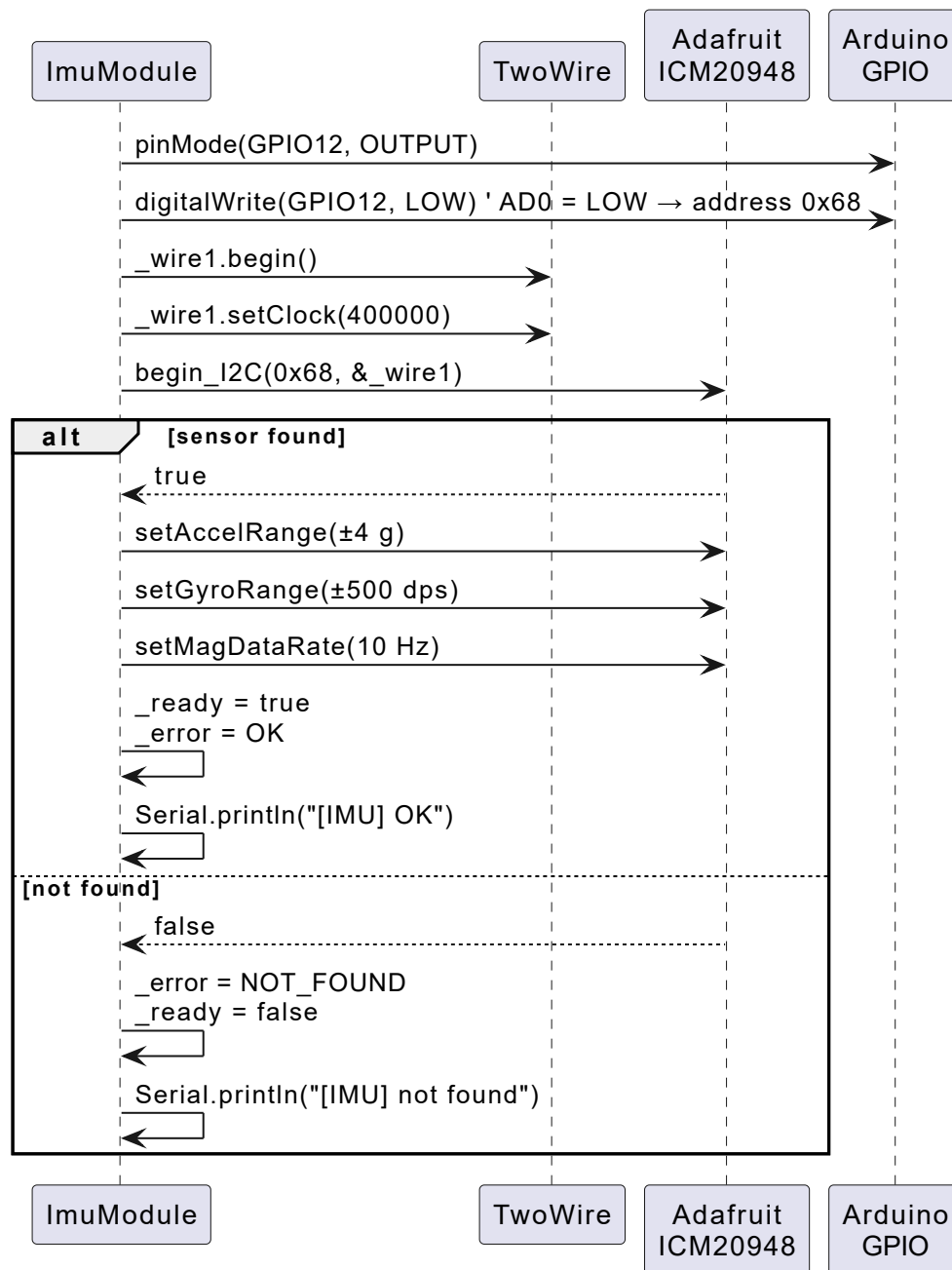
GPIO	Role	Direction
GPIO6	I2C1 SDA	bidirectional
GPIO7	I2C1 SCL	output (master)
GPIO12	AD0 address select	OUTPUT, always LOW

Implementation note: `Wire1` is not exported by the Arduino-Mbed variant for the Pico.
A dedicated instance is created: `static TwoWire _wire1(PIN_IMU_SDA, PIN_IMU_SCL);`

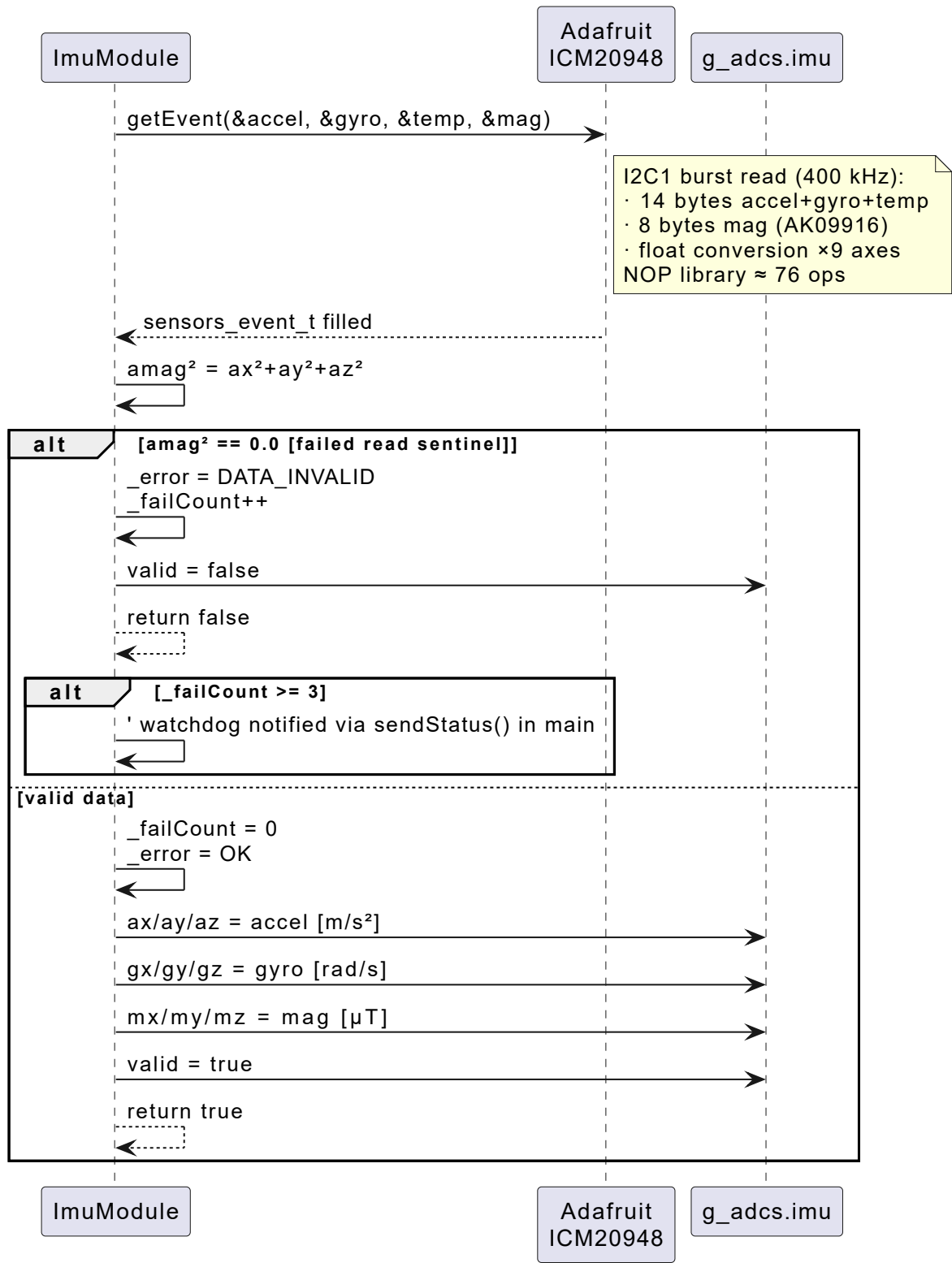
2. Class Diagram



3. Initialization Sequence — `init()`



4. Measurement Sequence — `read()`



5. Sensor Configuration

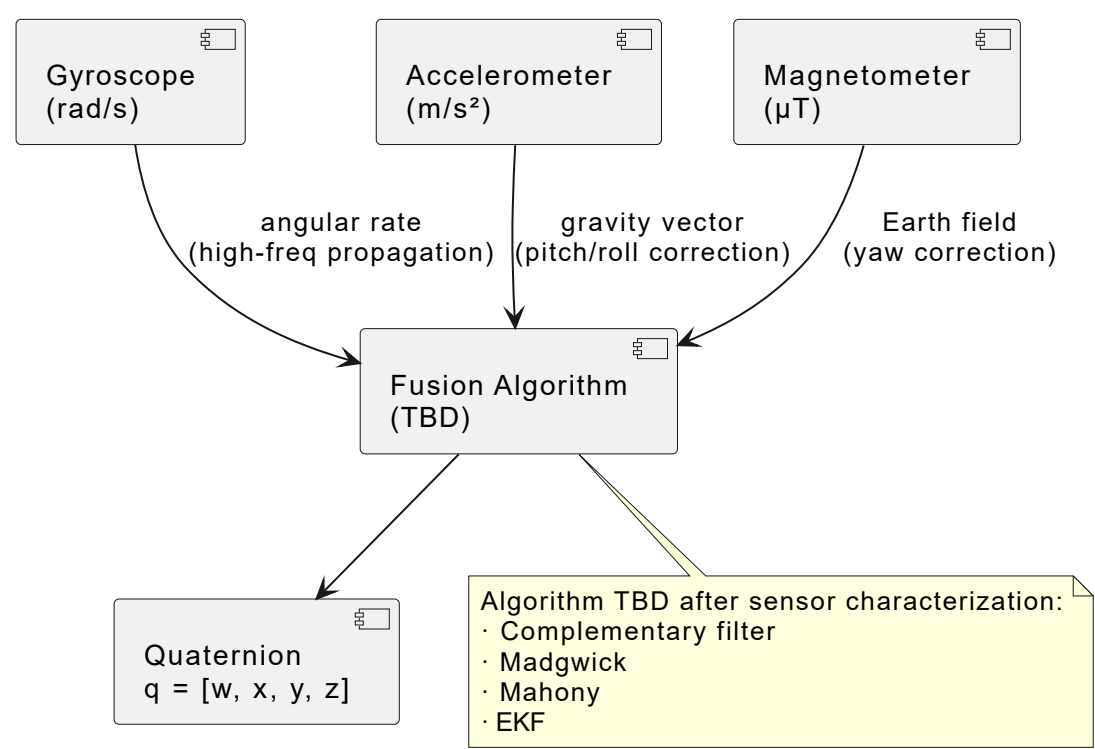
Parameter	Value	Register constant
Accelerometer range	±4 g	ICM20948_ACCEL_RANGE_4_G
Gyroscope range	±500 dps	ICM20948_GYRO_RANGE_500_DPS
Magnetometer rate	10 Hz	AK09916_MAG_DATARATE_10_HZ
I2C address	0x68	IMU_I2C_ADDR (config.h)

Parameter	Value	Register constant
I2C frequency	400 kHz	IMU_I2C_FREQ (config.h)

6. Output — g_adcs.imu

Field	Unit	Source axis
ax, ay, az	m/s ²	ICM-20948 accelerometer
gx, gy, gz	rad/s	ICM-20948 gyroscope
mx, my, mz	μT	AK09916 magnetometer
valid	bool	false if read fails or amag ² = 0

7. Attitude Fusion (future)



Quaternion avoids gimbal lock inherent to Euler angles.